Phage Infection Curves

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To determine how different **bacterial cell densities** influence growth, cell division, and morphology in *Xanthomonas citri* cultures.

Materials

- *Xanthomonas citri* subsp. *citri* strain
- NYG growth medium (peptone, yeast extract, glycerol)
- NYG agar plates
- Phosphate-buffered saline (PBS)
- Agarose (1%)
- 125 mL Erlenmeyer flasks
- Polypropylene tubes
- Spectrophotometer (OD600 measurement)
- Rotary shaker incubator (30°C)
- Pipettes and tips
- Microscope slides
- Fluorescence or phase-contrast microscope
- Petri dishes

Procedure

1. Culture Preparation

1. Streak *Xanthomonas citri* on an NYG agar plate.
2. Incubate at **30°C for 48 hours**.
3. Restreak isolated colonies onto fresh NYG agar plates.
4. Incubate another **48 hours** to obtain sufficient bacterial biomass.

2. Preparation of Bacterial Suspension

1. Transfer bacterial biomass into **50 mL NYG medium** in a polypropylene tube.
2. Measure the **optical density (OD600)** using a spectrophotometer.
3. Dilute cultures with fresh NYG medium to prepare **different starting densities**.

3. Growth Experiment

1. Transfer **50 mL of each density treatment** into separate **125 mL Erlenmeyer flasks.**
2. Incubate flasks at **30°C with shaking (200 rpm)**.
3. Maintain cultures under these density conditions for **6 hours**

4. Growth Measurement

1. Measure **OD600** every **6 hours** for up to **36 hours**.
2. Record growth curves for each density treatment.

5. Viability Testing

1. Perform **serial dilutions** of each culture.
2. … **agar plates**.

3. Incubate plates at **30°C for 48 hours**.
4. Count colonies to determine **CFU/mL (colony forming units per mL)**.

6. Microscopy Analysis

1. Place **20 µL of each culture** onto microscope slides coated with **1% agarose in PBS**.
2. Observe cells using **100× magnification**.
3. Measure **cell length and morphology** for at least **200 cells per treatment**.

7. Data Analysis

1. Compare:
    - Growth curves
    - CFU counts
    - Cell lengths
2. Perform **statistical analysis (Student's t-test)** to determine if density significantly affects cell division or morphology.